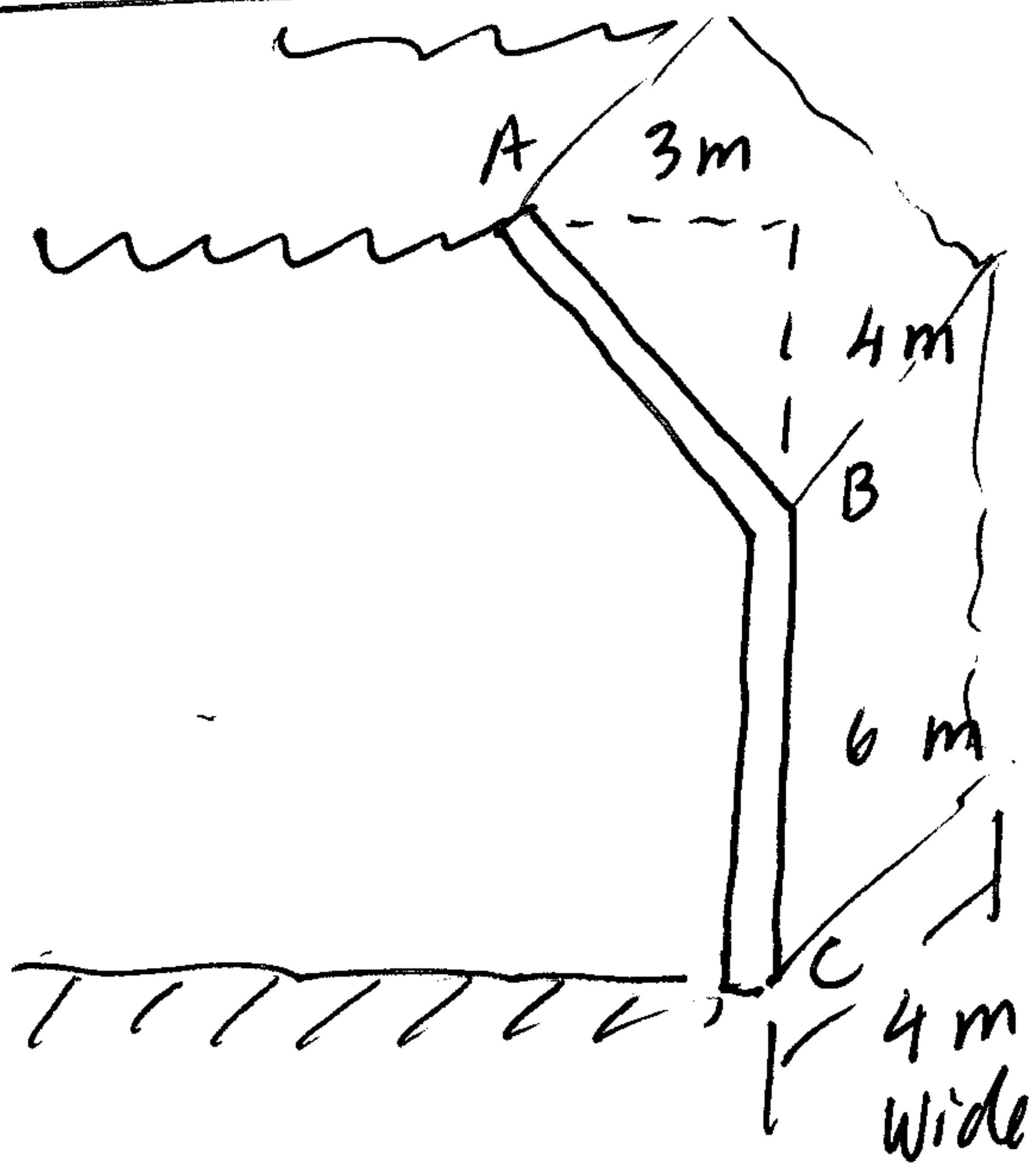


Another Fluid Pressure Problem worked 3 Ways



$$\gamma = 9810 \text{ N/m}^3$$

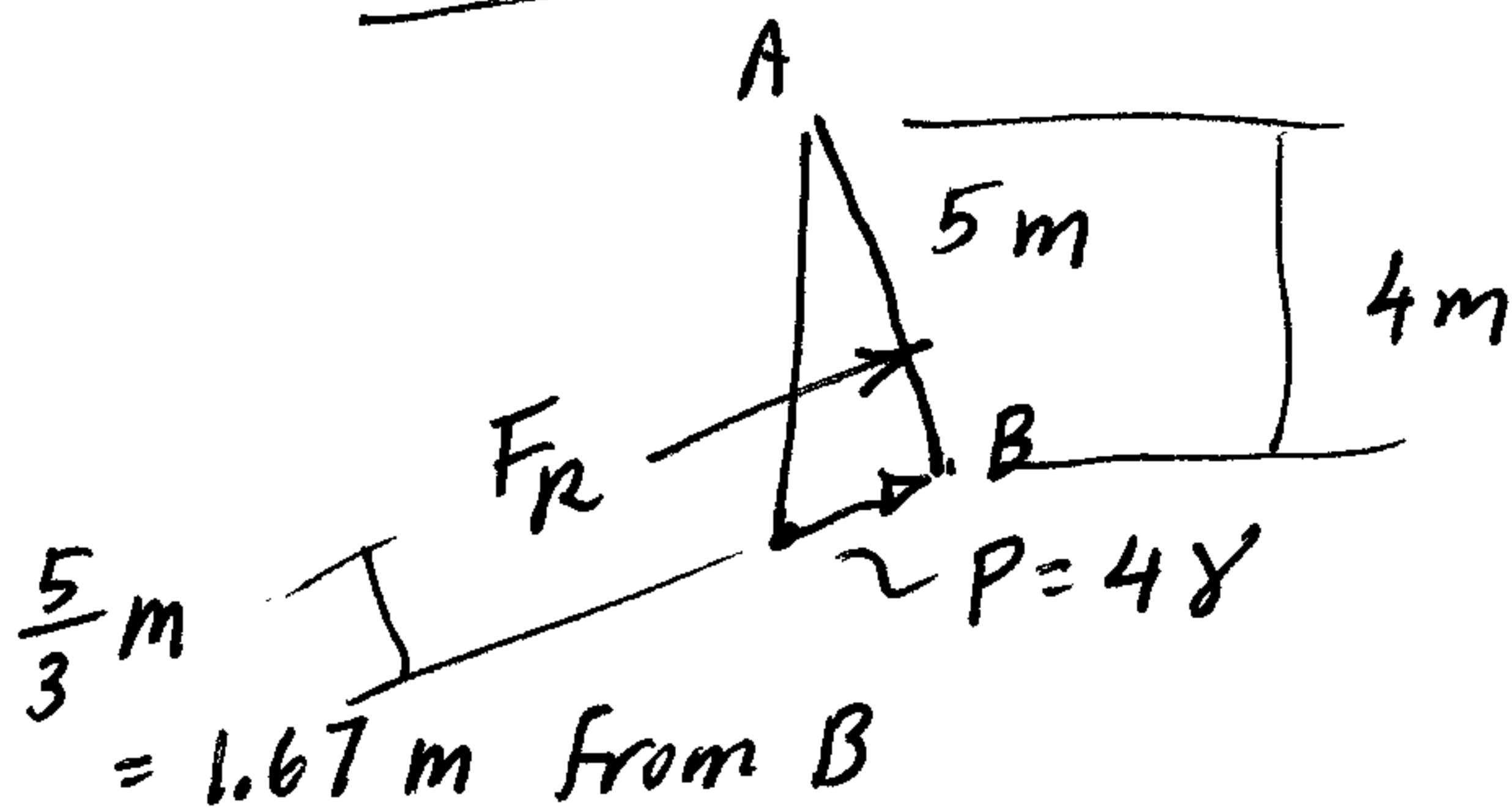
(Fresh Water)

Determine :

- ① F_R on AB
- ② F_R on BC

① F_R on AB

(a) Pressure Distribution

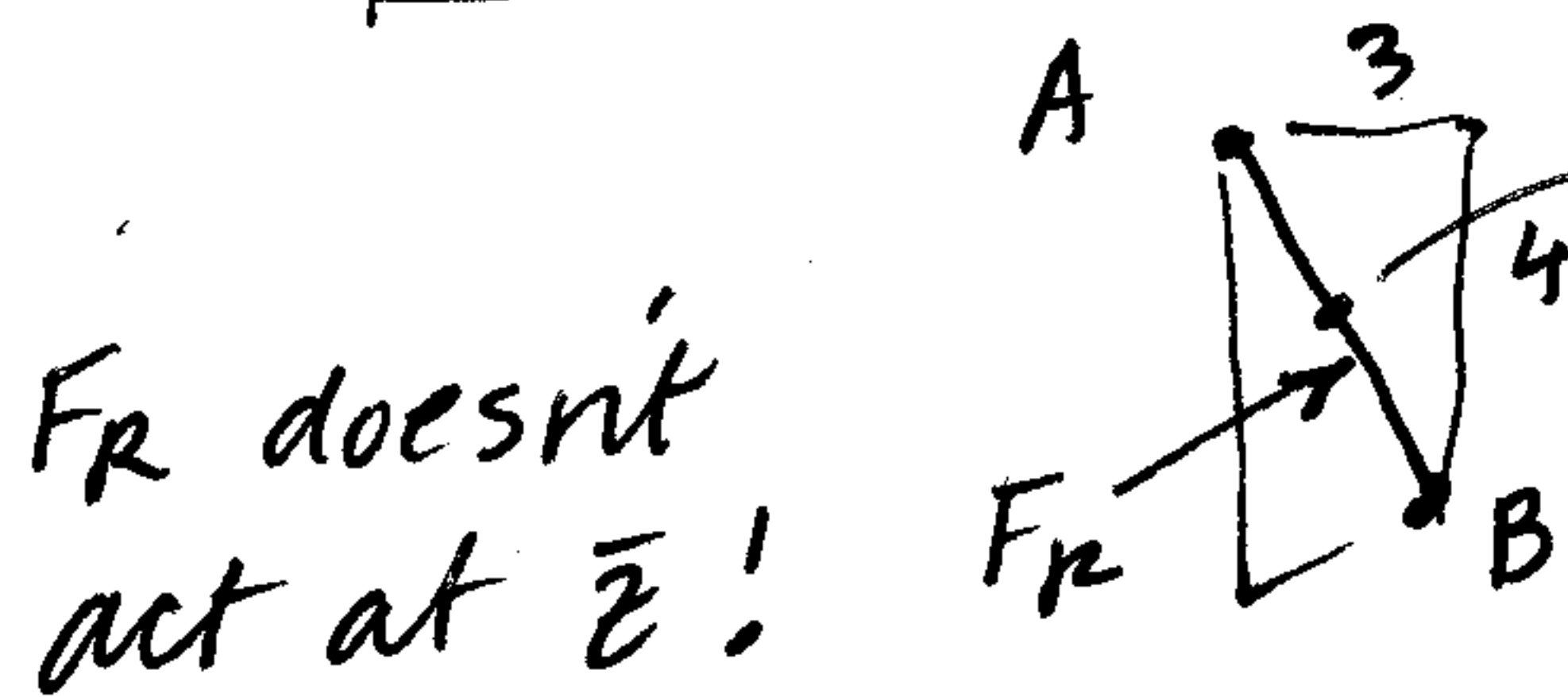


$$F_R = \frac{1}{2} b h \cdot w$$

$$= \frac{1}{2} (4\gamma) (5 \text{ m}) (4 \text{ m})$$

$$F_R = 392.4 \text{ kN}$$

(b) $F_R = \gamma \cdot \bar{z} \cdot A$



F_R doesn't
act at $\bar{z}$!

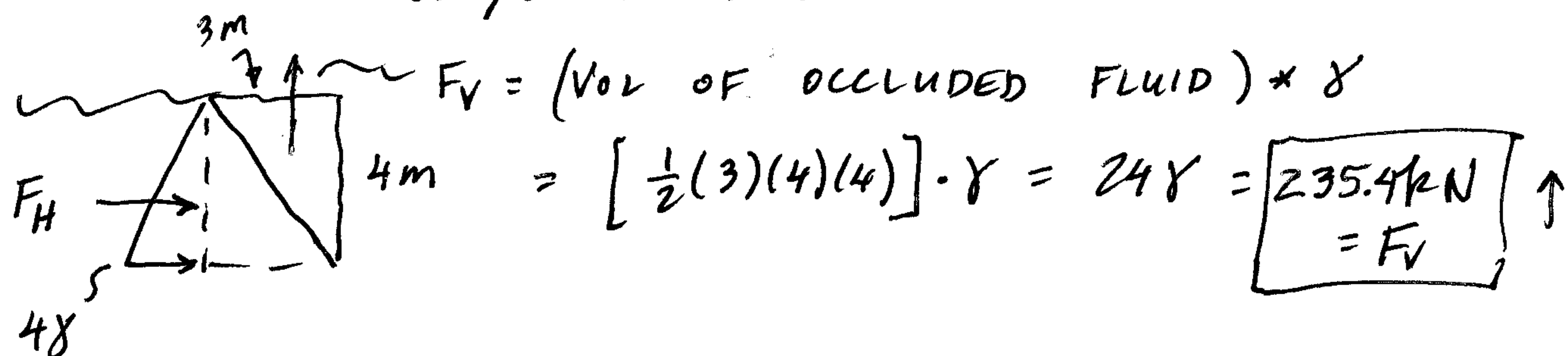
$$\bar{z} = \frac{1}{2} 4 = 2 \text{ m}$$

$$A = (5)(4) = 20 \text{ m}^2$$

$$F_R = (9810)(2)(20)$$

$$F_R = 392.4 \text{ kN}$$

F_R on AB : Horiz and Vertical
"Composite" Method



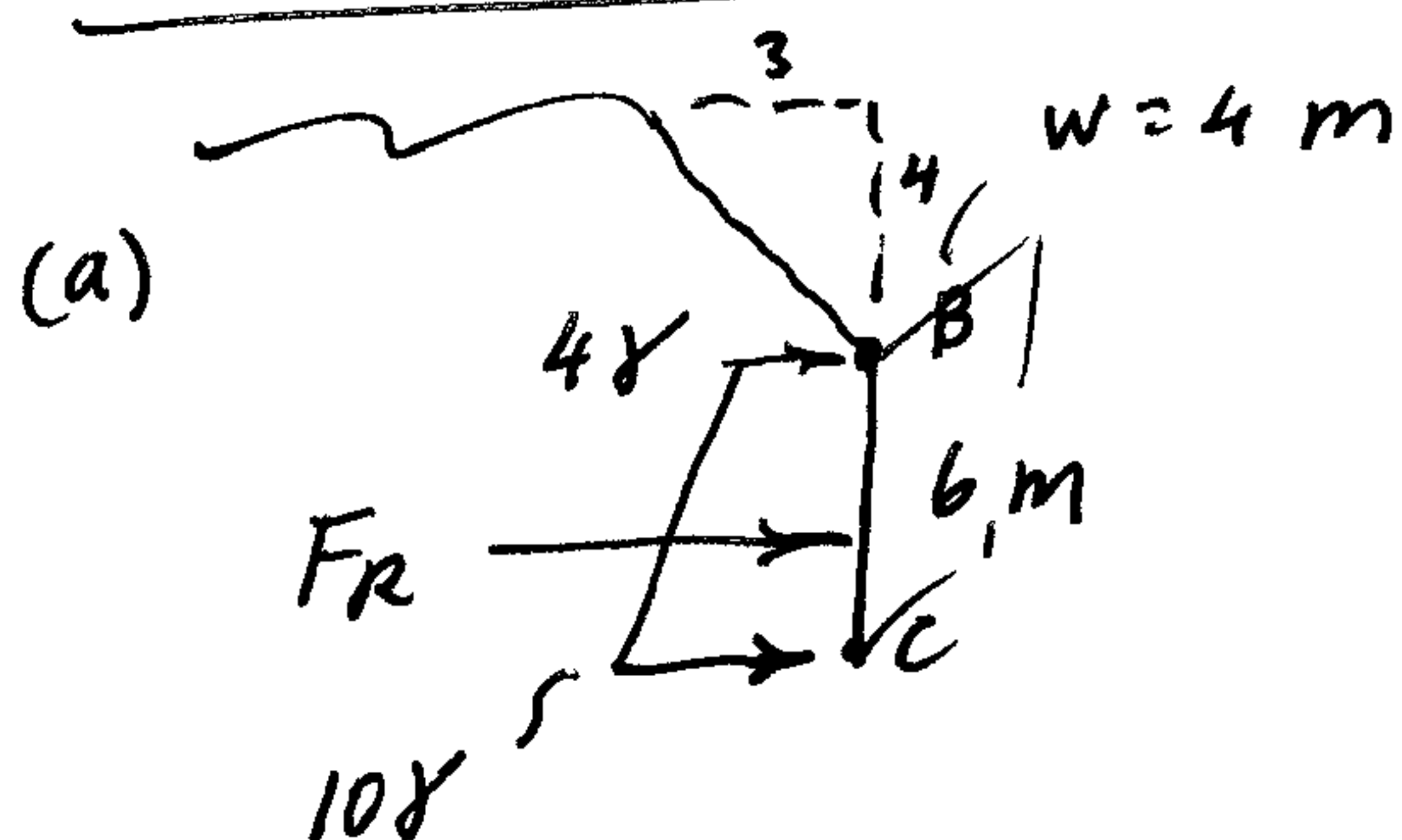
$$F_H = \frac{1}{2}bh \cdot w = \frac{1}{2}(4\gamma)(4)(4)$$

$$\boxed{F_H = 32\gamma = 313.9 \text{ kN} \rightarrow}$$

$$F_R = [313.9 \hat{i} + 235.4 \hat{j}] \text{ kN} = [392.4 \text{ kN} \angle 36.9^\circ]$$

$$\boxed{F_R = 392.4 \text{ kN} \nearrow \begin{smallmatrix} 3 \\ 4 \end{smallmatrix}}$$

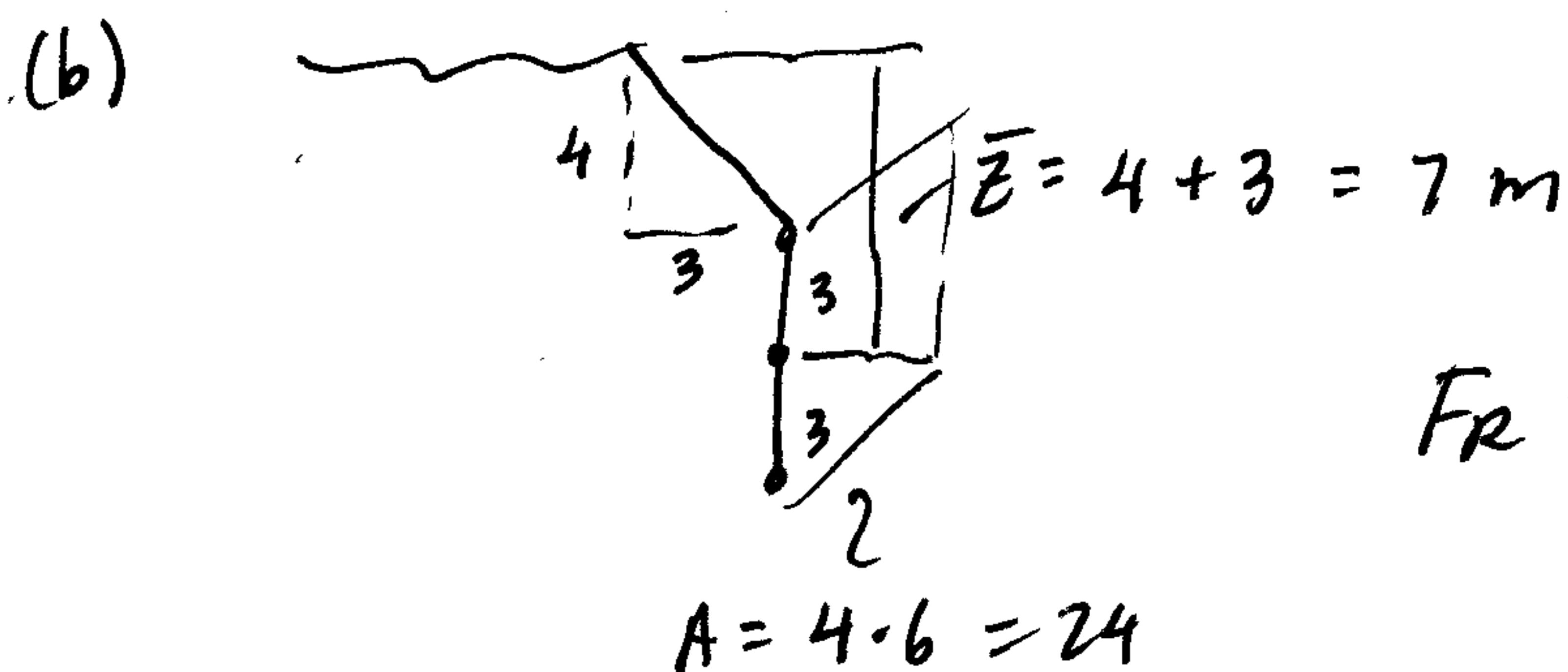
② F_R on BC



$$F_R = \left(\frac{4\gamma + 10\gamma}{2} \right) (6)(4)$$

$$F_R = 1648.1 \text{ kN} \rightarrow$$

$$= 1.648 \text{ MN}$$



$$\boxed{F_R = \gamma \cdot \bar{z} \cdot A}$$

$$F_R = (9810)(7 \text{ m})(24)$$

$$\boxed{F_R = 1.648 \text{ MN} \rightarrow}$$

(c) Cannot use $\vec{F}_R = \vec{F}_H + \vec{F}_V$ on this...
 $(F_V = 0)$ $\vec{F}_R = \vec{F}_H$ only...